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Aircraft joint with a curable composite bushing

Abstract

An aircraft structural joint with a first structural component provided with an attachment hole that is configured to receive an corresponding attachment fastener provided by a second structural component; a curable composite bush compressed and cured between an outer diameter of the attachment fastener and the attachment hole when the first component is mounted to the second component, wherein the curable composite bush, when cured, prevents radial displacement of the attachment fastener within the attachment hole.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 16/139,510, filed on Sep. 24, 2018, which claims the benefit of the Great Britain patent application No. 171970.6 filed on Nov. 28, 2017, the entire disclosures of which are incorporated herein by way of reference.

FIELD OF TECHNOLOGY

(1) The present technology relates to a curable composite bushing for an aircraft joint, an aircraft

joint, an aircraft structural assembly incorporating such an aircraft joint and a method and tool for installing a curable composite bushing.

BACKGROUND OF THE INVENTION

(2) During aircraft structural assembly, it is known to attach two or more components together at a joint by drilling and installing one or more fasteners in predetermined hole positions after the components are mounted at a fixed position relative one another. Once the fasteners are installed the joint is made and a larger structural assembly is formed. It is known that solid bushings or bushes can be use during the assembly of structural joints and assemblies. They are used when unintended eccentricity or gaps exist between a fastener and a corresponding hole into to which the fastener is placed, needs to be removed in order to ensure correct fit of the fastener. Normally a solid bush is machined from a blank or selected from a range of pre-machined solid bushes and then fitted with an interference fit within the existing bore. The solid bush may then be drilled in the desired position to provide a corrected bore into which the bolt is installed.

(3) In the assembly of aircraft structures, machining and drilling of such made to order solid bushing parts to the required accuracy is a precision process and therefore takes time and requires a stop in the assembly, therefore increasing overall costs of the assembly process.

(4) Sometimes the prepared solid bushing or the hole may not correspond to the dimensions of the hole exactly due to hidden irregularities in the surface of the hole itself, which results in a poor seating of the fastener when installed in the hole. Incorrect seating of the fastener in the hole results in the fastener not transferring the applied load fully when the aircraft is in operation, which has static and fatigue implications. Therefore, an aircraft structure may be designed with a conservative assumption that a certain number of fasteners in a structural assembly would be incorrectly seated fasteners. This conservative approach ultimately leads to a design with a higher number of fasteners required to transfer a given load, thus leading to a higher degree of redundancy, however the structural assembly will also be more costly to manufacture, heavier and require more maintenance due to the increased attachment part count. Furthermore, in some cases the hole may be so irregular that redundancy will not suffice for the degree of incorrect seating of the fastener, leading to the hole and the solid bushing needing to be reworked and or even the component to be scrapped. In addition, the use of solid bushings of various sizes requires storage and asset management, which takes up space of the assembly floor and resources. Furthermore, the process of fitting a solid bush into the existing bore may itself result in damage to the assembly, particularly in the case of composites. This again may require rework which can be time consuming and therefore can increase costs.

(5) An object of the technology is therefore to provide a bushing and an aircraft joint incorporating the bushing that is more efficient to build, cheaper to use and less likely to cause damage. A further object is to provide an aircraft structure assembly incorporating one or more joints with bushing in order to decrease assembly time of the structural assembly.

(6) A further object of the technology is to provide a tooling configured to install a bushing according to the object of the technology previously described. Lastly, it is an object of the present technology to provide a standard sized bushing that is an adaptable fit to multiple dimensions of a hole without machining, and furthermore, a standard bushing that is quicker to install and easier to store than those previously known.

BRIEF SUMMARY OF THE TECHNOLOGY

(7) According to an embodiment of the present technology, there is provided an aircraft structural joint comprising: a first structural component provided with an attachment hole that is configured to receive a corresponding attachment fastener provided by a second structural component; a curable composite bush compressed and cured between an outer diameter of the attachment fastener and the attachment hole when the first component is mounted to the second component, wherein the curable composite bush prevents radial displacement of the attachment fastener within the attachment hole. The curable composite bush may comprise a generally hollow cylindrical body

formed from a matrix material pre-impregnated with reinforcement fibers. The reinforcement fibers of the curable composite bush may be oriented in a generally circumferential direction about a longitudinal axis of the bush, wherein the body defines a plurality of corrugations extending between an inner and an outer diameter of the bush, and wherein the corrugations improve the compressibility of the bush in a direction substantially collinear to a longitudinal axis of the bush. The aircraft joint may further comprise a third structural component between the first and second structural components, wherein the third structural component is configured to displace the first structural component from the second structural component and is further provided with a hole configured to receive a portion of the protruding fastener element. One or more joints may be provided in an aircraft structural assembly. The aircraft structural may comprise a first component that is a leading-edge assembly and a second component that is a wing box spar element.

(8) A further embodiment of the present technology provides a tool for installing a composite bush comprising a compactor configured to compact the curable composite bush in a gap between an attachment fastener and an attachment hole such that the bush substantially conforms to the dimensions of the gap. The tool may further comprise one or more guides configured to engage an outer surface of the first structural component in proximity to the hole or the fastener and further configured to guide the curable composite bush into the gap. The guide may further be configured to guide the compactor into a compacting position. The tool may further comprise heating means configured to apply heat energy to the curable composite bush as it is guided or compacted. The heating means may be installed in the guide or the compactor. The heating means may be provided by an electric heating element, by a heated gas jet device, by an ultrasonic energy emitting device, or by an electrical inducting device.

(9) Another embodiment of the present technology provides a method of joining a first aircraft structural component to a second aircraft structural component, the method comprising the steps of: holding a first component provided with an attachment hole in an installation position relative to the second component provided with an attachment fastener, such that the attachment fastener is inserted in the attachment hole, which creates a gap between the fastener and the hole; placing a first end of a curable composite bush into the gap; providing a tool for installing the bush, the tool comprising one or more guides and a compactor; positioning a guide in proximity to the bush so that it encloses the bush within an inner surface; positioning a compactor within the guide and compacting the curable composite bush, at a further end of the compactor, into the gap between the attachment fastener and an attachment hole such that the bush substantially conforms to the dimensions of the gap so that no gap exists thereafter; holding the composite bush in the compressed state until it is cured. The method may further comprise the steps of heating the curable composite bush with heating means before, or during or after the compactor is moved towards the fastener. The method may further comprise the steps of providing a second guide and slidably engaging it the fastener so as to create a radial channel between the second guide and the compactor and feeding the curable composite bush into the channel. The method may further comprise the steps of moving the compactor into and out of engagement with the compressed curable composite bush to ensure the curable composite bush is sufficiently compressed into the gap.

(10) Advantages of the present technology will now become apparent from the detailed description with appropriate reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the technology will now be described, by way of example only, with reference to the following drawings in which:

- (2) FIG. 1 shows a plan schematic view of a commercial aircraft.
- (3) FIG. 2 shows a cross-section view through a wing structural assembly of FIG. 1 at the section A-A position shown, further showing a pair of identical aircraft joints.
- (4) FIG. 3 shows an isometric perspective view of a curable composite bush.
- (5) FIGS. 4A and 4B show schematic section views of embodiments of curable composite bush of FIG. 3, taken through the longitudinal axis of the bush, where the bush is in an uncompressed state.
- (6) FIG. 4C shows schematic section view of the embodiments of curable composite bush of FIGS. 3, 4A and 4B, taken through the longitudinal axis of the bush, where the bush is in a compressed state.
- (7) FIGS. 5A and 5B shows an aircraft joint of FIG. 2 at a first joining stage and a second joining stage, respectively, using a curable composite bush and an installation tool.
- (8) FIGS. 6A and 6B shows an aircraft joint of FIG. 2 at a first joining stage and a second joining stage, respectively, using a further type of curable composite bush and installation tool.
- (9) FIG. 7 shows a method of forming an aircraft joint of a structural assembly.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (10) With reference to FIG. 1, an aircraft **101** is shown with a wing **103** (also referred to as an airfoil) extending approximately horizontally through a fuselage **105**. A further wing **107** (also referred to as the horizontal tail plane) extends approximately horizontally from either side of a rear portion of the fuselage **105**. Yet a further wing **109** (also known as the vertical tail plane) extends vertically from an upper rear portion of the fuselage **105**.
- (11) The wings **103**, **109**, **107** and the fuselage **105** may be formed of structural assemblies joined together by structural joints which function to restrain at least a pair of assemblies from moving relative to one another once the joints have been fastened, in other words, the pair of assemblies are fixedly attached to one another. Each structural assembly may itself be formed of smaller structural assemblies joined together by a further number of joints.
- (12) The wings **103**, **107**, **109** each have a leading edge **111** and a trailing edge **113**, and a set of movable high-lift or control elements **115**, such as slats, flaps, ailerons, rudders and elevators, which are moveable (i.e., non-fixed) devices actuatable during operation between a deployed position and a retracted position according to the inputs of a controller. Adjacent, above or underneath each movable element **115** and/or in the areas where no movable elements **115** are provided, the leading edge and trailing edge structure of the wing **103** is fixed, i.e., not moving to function as a movable element **115**.
- (13) Geometrical characteristics of the aircraft **101** and its structural assemblies may be described with reference to a set of orthogonal principal aircraft axes X, Y and Z. The longitudinal axis (X) has its origin at the center of gravity **117** of the aircraft and extends lengthwise through the fuselage **105** from the nose to the tail in the normal direction of flight. The lateral axis (Y) also has its origin at the center of gravity **117** and extends substantially crosswise from wingtip to wing tip. The vertical or normal axis (Z) also has its origin at the center of gravity and passes vertically through the center of gravity. A further pair of local axes are defined with reference to aircraft principal axes X, Y, Z for any portion of a given structural assembly. In the present example, a set of local axes X', Y', Z' is defined at a fixed leading edge structural assembly **104** located in proximity to the leading edge **111** of the wing **103**. The first axis Y' lies in parallel to the principal XY plane at a swept angle (commonly referred to as the wing sweep angle) from the principal Y axis. This may be referred to as the spanwise axis. The second axis X' again lies in parallel to the same XY plane and is perpendicular to the Y' axis at the point of its origin, which can be selected at any spanwise position, but in this case at a lower portion of the fixed leading edge structural assembly **104**.
- (14) With reference to FIG. 2, and according to an embodiment of the present technology, a section view is shown of a fixed leading-edge assembly **104** at a spanwise A-A of FIG. 1 after completion of an assembly process.
- (15) The fixed leading-edge assembly **104** comprises a first structural component **201** that is joined

to a second structural component **205**. The first and second components are spanwise extending in the Y' dimension over substantially the full span of the wing **103** and are manufactured from composite CFRP material. At the position shown in FIG. **2** the components **201** and **205** are joined by a pair of joints **200A** (upper) and **200B** (lower); however, it should be appreciated that further joints substantially the same as **200A** and **200B** are similarly provided at spanwise intervals along the spanwise length of the first and second components **201**, **205**. In the present example, the first component **201** is a unitary, preassembled, structural module called a modular leading-edge assembly and the second component **205** is a unitary wing box front spar member **205**, however, it should also be appreciated that the components **201**, **205** may represent any plurality of structural components to be joined together to form any given structural assembly. It should also be appreciated that only one or more joints may be used for a given structural assembly, depending on the expected load transfer or failure design principle applied in the design of the assembly in question, e.g., safe-life or fail-safe (multiple load path) design principles. Preferably the types of joints **200A**, **200B** are used for highly loaded structural assemblies similar to the fixed leading edge structural assembly **104** shown, for example, between an attachment component for a movable, or at the joint between the wings **103,107,109** and the fuselage **105**.

(16) In the present embodiment, at the location of each joint **200A 200B**, the first component **201** is provided with an attachment hole **202** that is configured to receive a corresponding attachment fastener **203** provided by the second component **205**, the attachment fastener **203** having a longitudinal fastener axis **208**. Typically, the outer diameter Dfo of attachment fasteners **203** in such structures varies between 4 and 30 mm, however, in the present embodiment of a modular leading-edge assembly **104**, the diameter Dfo is approximately 10 mm.

(17) In the design of the assembly **104** and joint **200A/200B** shown, Dfo is smaller in dimension than the diameter Dh of the hole **202**, such that a first joining stage of the assembly process, when the first component **201** is first mounted to the second component **205**, a gap **501** (see FIG. 5A for more detail) exists between the fastener **203** and the hole **202**.

(18) Due to the design of the components **201**, **205**, but also due to the design of the manufacturing process, dimensions of the gap **501** may be exacerbated due eccentricity (e.g., the degree of non-circular dimension) of the hole **202** as a result of incorrect forming of the hole **202** during manufacture of the first component **201** or as a result of the hole **202** being formed an incorrect position in the first component **201** relative to the position of the fastener **203**. This is commonly referred to as an "out of tolerance" hole **202**. Typically, the cross-sectional dimension of the gap can be between 0.5 mm and 6 mm, depending on the size of the hole **202** and its eccentricity.

(19) In FIG. **2**, the gap **501** is not apparent because it is occupied completely by a curable composite bush **207**. The curable composite bush **207** is installed in a substantially pliable, uncured state between the hole **202** and the fastener **203** when the first and second components **201**, **205** are in a desired fixed installation position relative to one another, such that the bush **207** is able to comply with the surfaces of the hole **202** and the fastener **203** (regardless of how irregular their shape is) and fill the gap **501**.

(20) Once this is achieved, the curable composite bush **207** is then cured to an extent such that its material stiffness is increased so as to prevent radial displacement of the attachment fastener **203** within the attachment hole **202** and to provide optimum seating and load transfer between the fastener **203** and the first component **201** and second component **205**. A nut **211** and a washer **213** and are installed on the fastener **203** at each joint **200A** and **200B**, such that the washer **213** engages the surface of the first component **201** and the cured bush **207**. The nut **211** is threadably engaged and torqued so that a load bearing joint is formed between the first component **201** and the second component **205**.

(21) Use of a joint according to the present technology that incorporates a curable composite bush **207** in the way described allows for a structural assembly, and a manufacturing or assembly process that does not require the installation and machining of solid bushings, which is advantageous. It

also ensures that each structural fastener is fully load carrying, which may also permit less conservative static load and fatigue load design assumptions in the design of the assembly, leading to a lighter structural assembly design of lower part count or of lower manufacturing and maintenance cost per hole.

(22) Use of such joints **200A**, **200B**, may further enable structural design and assembly philosophies that are more efficient and more tolerant for gaps **501** that may exist between the attachment elements **202**, **203**, of structural components **201**, **205**. It should be appreciated that the structural design or assembly **104** may incorporate a joint **200A**, **200B**, according to the present technology by design, rather than as a remedy to incidental out of tolerance holes. This may be particularly beneficial to achieve a high rate manufacture of structural assemblies **104**, particularly those where the type of material used for the components **201**, **205**, the dimensions of components **201**, **205**, or the number of components **201**, **205** to be joined, inherently leads to a high variance and incidents of gaps **501** that need to be tolerated in the design of the joint **200A**, **200B**. For example, this is particularly preferred in the assembly of large modular leading-edge assemblies **104**, as described with reference to FIG. 2.

(23) Each joint **200A** **200B** further comprises a third structural component **209** in the form of a solid donut shaped spacer of equal thickness in the X direction and manufactured from CFRP composite material. The third component **209** may be manufactured using any suitably alternative material such as GFRP, polymer or metallic alloy material.

(24) At each joint **200A** **200B**, each third structural component **209** is configured to displace the first structural component **201** from the second structural component **205** in the $\pm X'$ direction. It may also be configured to reduce the contact surface area between first component **201** and second component **205** such that load transfer is substantially provided through the attachment fasteners **203**. It may be preferable that each component **209** may be machined or fettled to correct any misalignment between the first component **201** and the second component **205** in the direction mentioned. The third component **209** of each joint **200A** and **200B** may alternatively not be of the same dimension and that the third component **209** of each joint **200A** **200B** may be machined differently, in order to achieve a desired correction in alignment between the first component **201** and the second component **205**. As shown the third component **209** is provided with a hole **210** configured to receive a portion of the attachment fastener **203**, such that it is supported in a radial direction, i.e., perpendicular to the fastener axis **208**, however, such support may not be necessary.

(25) With reference to FIG. 3, an exemplary curable composite bush **207** for use in either of the joints **200A** and **200B** of FIG. 2 is shown. The bush **207** is shown in an uncured state. The bush **207** is formed by an open ended, elongate, corrugated and generally cylindrical body **301**. The corrugations **307** of the present embodiment are substantially chevron shaped. The body **301** further has a central longitudinal axis **304** that defines a center point for cross sections of the body **301** taken perpendicularly along the longitudinal axis **304**. The body **301** is formed from a composite material composed of reinforcement material **303** pre-impregnated with a matrix material **302**.

(26) With reference to FIG. 4A, a cross-section of the body **301** is shown. The cross-section lies on a plane formed by the longitudinal axis **304** and a line perpendicular to the axis **304**.

(27) The body **301** has a thickness t that is constant along its length $L1$. The thickness of the present embodiment is approximately 0.5 mm. The reinforcement material **303** is formed of continuous fibers that are evenly distributed through the thickness t of the body **301** and along the length $L1$ of the body **301**. The fibers are orientated substantially concentrically about the longitudinal axis **304** of the body **301**, and are generally aligned in parallel with a circumference **305** of the body **301** at any cross-section of the body **301** viewed on a plane that is perpendicular to the axis **304**.

(28) A pair of exemplary magnified views **V1** and **V2** is provided in order to demonstrate the distribution of the matrix material **302** and reinforcement material **303** along the length and

thickness t of the body **301**, as well as the orientation of the reinforcement material **302**. The ends of the fibers can be seen as dots **303**, which is representative of their orientation.

(29) The body **301** is of continuous cross-section with constant dimension values of D_o and D_i , respectively. However, the body **301** may vary in cross-sectional dimensions along its length. For example, the body **301** may be tapered from one end to the other end. The cross-section used may alternatively be elliptical, square, triangular or any combination thereof, as required to suit the characteristics of the attachment hole.

(30) Chevron shaped corrugations **307** define the inner diameter D_i and the outer diameter D_o of the bush **207** and are configured allow the bush **207** to be compressed and allow the body **301** to collapse in the direction of the longitudinal axis **304** such that the corrugations **307** overlap in series, when the bush **207** is compressed along the axis **304**. D_i is dimensioned to substantially the same dimension of D_{fo} of the attachment fastener **203**, to which the bush **207** is to be applied. Chevron shaped corrugations are preferable as they permit the highest packing density and most even distribution of fibers both radially from, and in parallel to, the axis **304** when the bush **207** is brought into a compressed state, thus ensuring a more even mechanical stiffness property through the bush **207**, when it is installed and cured in an attachment hole.

(31) With reference to FIG. **4B**, an alternative embodiment of the present technology is shown. A bush **207** substantially in accordance with the previous embodiment comprises the same length L_1 , thickness t , material types and inner and outer diameters, D_i , and D_o . However, in the present embodiment, the corrugations **307** have a sinusoidal profile. This may be preferable over alternative shapes such as chevron shaped corrugations **307**, because sinusoidal corrugations **307** may be less prone to point damages and may be easier to manufacture, even though they may result in a less evenly distributed fibers when the bush **207** is brought into a compressed state.

(32) Furthermore, the corrugations **307** are defined continuously at a non-perpendicular angle Φ to the longitudinal axis, such that the corrugations **307** extend in the form of a helix along the length L_1 of the curable composite bush **207**. This helix may equally be provided for the previously described chevron shaped corrugation or for any suitable alternative corrugation shape that is capable of being provided in a helical 3D form. The helical form may be preferable to enable the bush **207** to be manufactured using a continuous extrusion process.

(33) With reference to FIG. **4C**; a cross-section is shown of the curable composite bush **207** of FIGS. **4A** and **4B** in a compressed and cured state, which is the state of the bush **207** that fills the gap shown in FIG. **2**. In the cured state of FIG. **2**, the axis **208** of the fastener aligns concentrically to the axis **304** of the bush **207**, such that the values of D_{fo} and D_i are substantially the same. The corrugations **307** substantially overlap in series when the bush **207** is compressed to a compressed length L_2 , such that D_o and D_h are also the substantially same dimension. A half-length l , defines the radial dimension of overlap with respect to the longitudinal axis **304** of the bush, which in the present example is 3 mm.

(34) The matrix material **302** used in the bush **207** of FIGS. **3**, **4A** and **4B** is Polyetheretherketone (PEEK); a thermoplastic matrix material, however any other suitable thermoplastic matrix material may alternatively be used such as polyethersulfide (PES), polyetherimide (PEI), or polyphenylenesulfide (PPS). For the purpose of this description, a curable composite bush **207** formed in part from a thermoplastic matrix material **302** is said to be in a “cured” state, when the temperature of the thermoplastic matrix material **302** is brought below its applicable melting temperature and the bush **207** is in a compressed state. It is said to be in an “uncured” state when the temperature of the thermoplastic matrix material **302** has reached or is above its applicable melting temperature and/or the bush **207** is in a decompressed or uncompressed state.

(35) The body **301** may alternatively be formed from a reinforcement material **303** pre-impregnated with a partially or non-polymerized thermosetting matrix material **302**, chosen from one of matrix materials commonly used in aerospace such as polyester, epoxy, vinylester, bismaleimide, phenolic or polyimide. For the purpose of this description, a curable composite bush **207** formed in part

from a thermoset matrix material **302** is said to be in a “cured” state, when the bush **207** is deformed in a compressed state and the matrix material **302** polymerized such that bush **207** is irreversibly deformed.

(36) The reinforcement material **303** in the present embodiment is composed of continuous glass fibers of alumina borosilicate glass otherwise known as ‘E-glass’, however any other suitable continuous glass fiber reinforcement may be used, for example S-Glass. In addition, a reinforcement material **303** using other material types may be used such as graphite/carbon type fibers or aramid type of fibers. Continuous fibers are preferable because during compression of the bush **207** their length ensures that they remain oriented substantially concentrically about the longitudinal axis **304** of the body **301** in parallel with a circumference **305** of the body **301**, which would not be the case for short fibers, which may re-orientate and ultimately hinder the collapse of the bush **207** in a desired way. Furthermore, once compressed within the gap **501**, the continuous fibers will remain generally aligned and will interact so that composite material is evenly distributed between the hole **202** and fastener **203** when compressed.

(37) Use of a thermoplastic matrix material **302** may be preferable as it allows the curable composite bush **207** to be handled, stored more easily during assembly operations when the matrix material of the bush **207** is below its applicable melting temperature, as shown in FIGS. **3** and **4A**, and is resiliently deformable. In this state, the bush **207** is less prone to unintentional damage, and can be handled more easily, than a curable composite bush **207** with a body **301** formed from a reinforced, partially cured thermoset resin matrix material **302**, for example.

(38) It may also be advantageous to use a thermoplastic matrix material **302** as it does not require special storage and shelf life considerations that could be necessary were the body **301** to be formed using a partially cured thermoset resin matrix material **302**, which have a pre-determined shelf life and may require cooling to provide a usable shelf-life in an assembly line

(39) Lastly, disassembly of the joint **200A**, **200B** may be required at some point in the aircraft's life cycle, and it is foreseen that removal of the bush **207** by application of heat is easier than application of machining, which would be required for removing a thermosetting type of bush **207**.

(40) That said, use of a thermosetting matrix material **302** may in some instances be preferable, particularly where a matrix material with higher mechanical performance properties is required or where a matrix material **303** with irreversible properties is required due to a high temperature environment of the joint **200**.

(41) The use of E-glass fibers as a matrix material **302** may be advantageous as they have higher compression strength properties when compared to alternative high-performance fibers, which may alternatively be used in aircraft structure, for example carbon fiber or aramid fiber. Furthermore, glass fibers are galvanically compatible with a wider range of structural materials commonly used in aircraft joints, for example, titanium or aluminum alloy.

(42) It should be appreciated, that the volume ratio of reinforcement material **303** to matrix material **302** may be varied depending on the specific mechanical stiffness and strength properties required from the bush **207** when it is compressed and cured in a joint **200A**, **200B**. Furthermore, mixtures of different type of reinforcement material **303** including fibers may be used, if required. An uneven distribution of the reinforcement material **303** may also be used and tailored to suit the principal load direction and levels between the attachment fastener **203** and attachment hole **202**.

(43) With reference to FIGS. **5A** and **5B**, attachment of a first component **201** to a second component **205** is shown using a curable composite bush **207**.

(44) In FIG. **5A**, a first joining stage shows the first component **201** held in an installation position relative to the second component **205**, such that the attachment fastener **203** is inserted in the attachment hole **202**, which creates a gap **501** between the fastener **203** and the hole **202**. A first end **303A** of a curable composite bush **207** in accordance with the bush described in FIGS. **3**, **4A** and **4B**, is then introduced into the gap **501** between the outer diameter of the fastener D_{fo} and the diameter of the hole D_h , i.e., it is inserted over the exposed end of the fastener **203**

(45) A tool **500** for installing the bush **207** comprising a guide **503** and a compactor **505** is positioned in proximity to the bush **207**. The guide **503** and compactor **505** each comprise a body formed from steel and shaped as an open-ended cylinder. An end surface **504**, defined by the guide **503**, is substantially planar and has an inner diameter substantially the same as the diameter D_h of the hole **202**. The end surface **504** is configured to comply with an exterior surface of the first component **201** in proximity to the hole **202** such that the guide **503** can be steadily held in position by a user.

(46) An inner surface **506** of the guide **503** is configured to be substantially smooth and polished and may comprise a non-stick treatment. The inner surface **506** is configured so as to contact the bush **207** in order to ensure that it stays substantially cylindrical in form and orientated in the direction of its longitudinal axis **304** as the bush **207** is compressed into the gap **501**. The inner surface **506** of the guide **503** is further configured to guide the compactor **505** in a direction substantially parallel to the longitudinal axis **304** of the bush **207**.

(47) A further end surface **508**, defined by the compactor **505**, is substantially planar and has an inner diameter substantially the same as the diameter D_{fo} of the fastener **203** and an outer diameter substantially the same as the diameter D_h of the hole **202**. The compactor **505** is configured to slide within the guide **503** and to compact the curable composite bush **207**, at the further end **508**, into the gap **501** between the attachment fastener **203** and an attachment hole **202**, such that the bush **207** substantially conforms to the dimensions of the gap **501**. The use of the compactor **505** helps to ensure that compression pressure is evenly applied by the compactor **505** at its further end surface **508** to the bush **207**. The combination of the presently described compactor **505** and guide **503** is also advantageous as the relative displacement of the compactor **505** relative to the guide **503** may be used to determine the degree of compression of the bush **207** within the gap **501**. As such the amount of bush material compacted within the gap **501** can be derived from such measurements.

(48) In the present example, the curable composite bush **207** is formed partially of thermoplastic material **302**, therefore heating means are provided in the form of electrical heating elements **507** embedded with the guide **503** and separately with the compactor **505**. Electrical heating elements **507** may be preferable as they are easier to control than other means, such as liquid heating. Embedding the heating element **507** avoids any interference between the smooth inner surface **506** of the guide **503** and the bush **207** and reduces the likelihood of collecting contaminants that may be transferred between the tool **500** and the bush **207**. Such contaminants are undesirable as they may affect the curing of the bush **207** and may also pose a fire risk when heating is applied.

(49) It may alternatively be sufficient to attach the heating elements **507** to a surface portion of the guide **503** or compactor **505**. The heating means **507** may be activated once the bush **207** is positioned as shown within the gap **501**, before compacting of the bush **207** is started. Heating means **507** may only need to be provided in the guide **503** or the compactor **505**.

(50) Once the bush **207**, guide **503** and compactor **505** are put into the position shown in FIG. 5A, the compactor **505** is moved towards the fastener **203** to compress the bush **207** into the gap **501** in a second joining stage, so that no gap **501** exists thereafter, as shown in FIG. 5B. The curable composite bush **207** is then held in the compressed state shown in FIG. 5B until it is cured. The tooling **500** is then removed. In the cured state, the bush **207** prevents radial displacement of the attachment fastener **203** within the attachment hole **202**. In the case of installing a bush **207** partially formed from a thermoplastic matrix material **302**, the bush **207** is heated by the heating means **507** before the compactor **505** is moved towards the fastener **203** into order to bring the curable composite bush **207** into a melted pliable state, after which the compactor **505** can be moved to compress the bush **207** into the gap **501**. In the case of installing a bush **207** partially formed from a thermosetting matrix material **302**, heating means **507** may also be applied in order to accelerate the polymerization of the matrix material, which may be desirable in high joining rate applications.

(51) It should be appreciated for the embodiments so far described that, for certain applications, the

corrugations **307** may not be required and that the material properties of the bush, particularly its pliability, in response to heat application may suffice for feeding a bush into a gap **501**. With reference to FIGS. **6A** and **6B**, an exemplary curable composite bush **601** without corrugations is provided as well as a tool **600** for installing it. The joint **200A/200B** and tool **600** shown in FIGS. **6A** and **6B** are substantially in accordance with the previous embodiment shown in FIGS. **5A** and **5B**, however, the bush **601** is different in form than previously provided and the tool **600** has an additional second guide **603** fitted with a deployable radial cutter **605**.

(52) The curable composite bush **601** comprises a continuous tube of substantially circular cross-section and has a body **301** in the form of an un-corrugated cylinder. The body **301** of the bush **601** is composed of a thermoplastic matrix material **302** pre-impregnated with continuous fibers of a reinforcement material **303**, that are orientated about the longitudinal axis **304** of the body **301**, and are generally aligned in parallel with a circumference **305** of the body **301** at any cross-section of the body **301** when viewed on a plane that is perpendicular to the axis **304**.

(53) In FIG. **6A**, in a first joining stage the first component **201** is shown held in an installation position relative to the second component **205**, such that the attachment fastener **203** is inserted in the attachment hole **202**, which creates a gap **501** between the fastener **203** and the hole **202**. The second guide **603** is slidably engaged to the fastener **203**. The second guide **603** is cylindrical with an outer diameter at an end furthest from the fastener hole **203** that has a diameter smaller than inner diameter of the compactor **505**, such that a radial channel **607** exists between the second guide **603** and the compactor **505**. The outer diameter of the second guide **603** reduces towards the attachment end to the fastener **203** such that a taper is provided. The second guide is formed from steel, has a smooth, polished and non-stick surface and is provided with heating means **507** in the form of a single embedded heating element **507**. Once the second guide **603** is attached to the fastener, the guide **503** and compactor **505** are then positioned. Then, a first end of a curable composite bush **601** as described is fed around the second guide **603** and into the channel **607**.

(54) With reference to FIG. **6B**, once the bush **601**, guide **503** and compactor **505**, and second guide **603** are put into position shown in FIG. **6A**, a second joining stage ensues wherein the bush **601** is heated to a temperature approximately equal to or above the melting temperature of the thermoplastic material. Simultaneously or thereafter, the bush **601** is fed between the outer diameter of the fastener D_{fo} and the diameter of the hole D_h in the channel **607** towards the fastener **203** and into the gap **501**, until sufficient material is introduced to thereafter remove the gap **501**. The compactor **505** may then be moved into and out of engagement with the compressed curable composite bush **601** to ensure it has been sufficiently compressed into the gap **501**. The bush **601** is then left to “cure,” e.g., cool below its melting point in the compressed state, such that in the cured state shown in FIG. **6B**, the bush **601** prevents radial displacement of the attachment fastener **203** within the attachment hole **202**. Once the bush **601** cures, the compactor **505** is withdrawn away from the fastener **203** and the radial cutter is deployed to cut the unused portion of bush **601**, before the tool **600** is then removed.

(55) With reference to FIGS. **5A**, **5B**, **6A** and **6B**, it should be appreciated that the heating means **507** for the tooling **500**, **600** may be provided by other suitable means. A heating means **507** may instead be provided by a heated gas jet device attached at the othermost end of the guide **503** may be used. A heated gas jet device may be preferable as it may use less energy and provide a better heat distribution to the curable composite bush **207**. Alternatively, heating means **507** may be provided by an ultrasonic energy emitting device attached to the tool. Use of an ultrasonic energy emitting device may be used to direct ultrasonic wave energy to heat the material of the bush **207** and remove imperfections within it, when the material of the bush **207** is in a more pliable state.

(56) Alternatively, heating means **507** may be provided by electrical induction of the fastener **203** or fibers of the reinforcement material **303**, using an electrical inducting device. Such an alternative embodiment may be preferable where access to the bush **207** is particularly problematic.

(57) With reference to FIG. **7**, a method of joining a first aircraft structural component to a second

aircraft structural component is provided, the method comprising the steps of: **701**—holding first component **201** provided with an attachment hole **202** in an installation position relative to the second component **205** provided with an attachment fastener **203**, such that the attachment fastener **203** is inserted in the attachment hole **202**, which creates a gap **501** between the fastener **203** and the hole **202**; **703**—placing a first end of a curable composite bush **207** into the gap **501**; **705**—providing a tool **500** for installing the bush **207**, the tool **500** comprising one or more guides **503**, **603** and a compactor **505**; **707**—positioning a guide **503** in proximity to the bush **207** so that it encloses the bush **207** within an inner surface **506**; **709**—positioning a compactor **505** within the guide **503** and compacting the curable composite bush **207**, at a further end **508** of the compactor **505**, into the gap **501** between the attachment fastener **203** and an attachment hole **202** such that the bush **207** substantially conforms to the dimensions of the gap **501** so that no gap **501** exists thereafter; **711**—holding the composite bush **207** in the compressed state until it is cured.

(58) The method may further comprises the steps of: **713**—heating the curable composite bush **207** with heating means **507** before, or during or after the compactor **505** is moved towards the fastener **203**; **715**—moving the compactor **505** into and out of engagement with the compressed curable composite bush **207** to ensure the curable composite bush **601** is sufficiently compressed into the gap **501**; **717**—providing a second guide **603** and slidably engaging it a fastener **203** so as to create a radial channel **607** between the second guide **603** and the compactor **505**, into which a curable composite bush **601** may be fed; and **719**—engaging a radial cutter to remove any excess curable composite bush **601**.

(59) Where in the foregoing description, integers or members are mentioned which have known, obvious or foreseeable equivalents; then such equivalents are herein incorporated as if individually set forth. Reference should be made to the claims for determining the true scope of the present technology, which should be construed so as to encompass any such equivalents. It will also be appreciated by the reader that integers or features of the technology that are described as preferable, advantageous, convenient or the like are optional and do not limit the scope of the independent claims. Moreover, it is to be understood that such optional integers or features, whilst of possible benefit in some embodiments of the technology, may not be desirable, and may therefore be absent, in other embodiments.

(60) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. A tool for installing a composite bush, the tool comprising: a compactor configured to compact a curable composite bush in a gap between an attachment fastener and an attachment hole such that the bush substantially conforms to dimensions of the gap, one or more guides configured to engage an outer surface of a first structural component in proximity to the hole or the fastener and further configured to guide the curable composite bush into the gap, and a heating means configured to apply heat energy to the curable composite bush as it is guided or compacted.

2. The tool according to claim 1, wherein the wherein the curable composite bush comprises a hollow cylindrical body formed from a matrix material pre-impregnated with reinforcement fibers,

and wherein the reinforcement fibers of the curable composite bush are oriented in a circumferential direction about a longitudinal axis of the curable composite bush, wherein the hollow cylindrical body defines a plurality of corrugations extending between an inner and an outer diameter of the curable composite bush, and wherein the corrugations improve a compressibility of the curable composite bush in a direction collinear to the longitudinal axis of the curable composite bush.

3. The tool according to claim 1, wherein the one or more guides are further configured to guide the compactor into a compacting position.

4. The tool according to claim 1, wherein the heating means is installed in the one or more guides or the compactor.

5. The tool according to claim 1, wherein the heating means is provided by an electric heating element.

6. The tool according to claim 1, wherein the heating means is provided by a heated gas jet device.

7. The tool according to claim 1, wherein the heating means is provided by an ultrasonic energy emitting device.

8. The tool according to claim 1, wherein the heating means is provided by an electrical inducting device.
